

ComoRAG – Next-Gen RAG for Long-Document Reasoning - Demo



Fahd Mirza
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See how ComoRAG brings cognition-inspired, multi-step retrieval to handle huge documents and complex stories using OpenAI models.

https://github.com/EternityJune25/ComoRAG

README MIT license

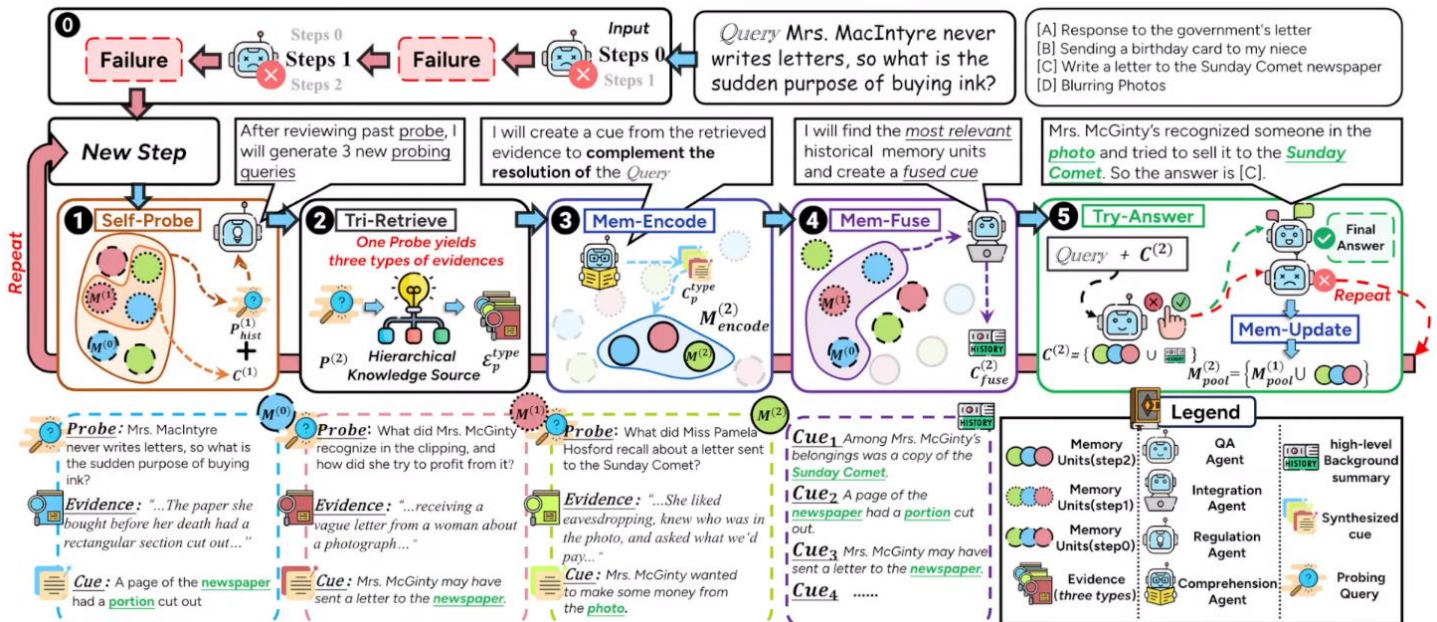
ComoRAG: A Cognitive-Inspired Memory-Organized RAG for Stateful Long Narrative Reasoning

Python 3.10+ CUDA 12.x Platform Linux RAG Graph Memory LLM OpenAI/vLLM Status Active arXiv 2508.10419

License MIT DeepWiki ComoRAG

English | 中文

The diagram illustrates the ComoRAG architecture, which is a cognitive-inspired memory-organized RAG for stateful long narrative reasoning. It shows a multi-step process starting from an input query: "Query Mrs. MacIntyre never writes letters, so what is the sudden purpose of buying ink?". The process involves several steps: 1. Self-Probe, 2. Tri-Retrieve, 3. Mem-Encode, 4. Mem-Fuse, and 5. Try-Answer. Each step is represented by a box with a number and a description of the action. The process is iterative, with a "Repeat" loop indicated by a red arrow. The final output is the "Final Answer" [C].



← → ↻ https://github.com/EternityJune25/ComoRAG

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Key Features 🌟

- Support for various LLMs and local/remote embedding models
- Graph-augmented retrieval and reasoning
- 🔧 Flexible data preprocessing and chunking
- 📊 Multiple evaluation metrics (F1, EM, etc.)
- Modular and extensible design

Directory Structure 📁

```
ComoRAG/
├── main_openai.py          # Main program using OpenAI API
├── main_vllm.py            # Main program using local vLLM server
├── script/                 # Data processing and evaluation scripts
│   ├── chunk_doc_corpus.py # Document chunking script
│   └── eval_qa.py          # QA evaluation script
├── dataset/                # Dataset directory
│   └── ...
├── src/comorag/            # Core code
│   ├── ComoRAG.py         # Main class and core logic
│   ├── utils/             # Utility modules
│   ├── embedding_model/    # Embedding model related
│   ├── llm/               # LLM related
│   ├── prompts/           # Prompt templates
│   ├── information_extraction/ # Information extraction
│   └── rerank.py, embedding_store.py # Other core modules
├── requirements.txt        # Dependencies
└── README.md              # Project documentation
```

← → ↻ https://github.com/EternityJune25/ComoRAG

README MIT license

Installation & Environment □

- 🐍 Python version: Python 3.10 or above recommended
- 📦 Install dependencies:

```
pip install -r requirements.txt
```
- 💡 Environment variables: Set your OpenAI API Key or local LLM/embedding paths as needed
- 🖨 GPU (optional but recommended): CUDA 12.x supported by many dependencies in requirements.txt

Data Preparation & Format ■

- 📄 Corpus file corpus.jsonl: Each line is a document, with fields like `id`, `doc_id`, `title`, `contents`
- ❓ QA file qas.jsonl: Each line is a question, with fields like `id`, `question`, `golden_answers`

Example:

corpus.jsonl:

```
{"id": 0, "doc_id": 1, "title": "...", "contents": "..."}

qas.jsonl:
```

```
Ubuntu@0360-kci2-ty6k-prxmx100037:~$ conda create -n ai python=3.10 -y && conda activate ai
Channels:
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): \
```

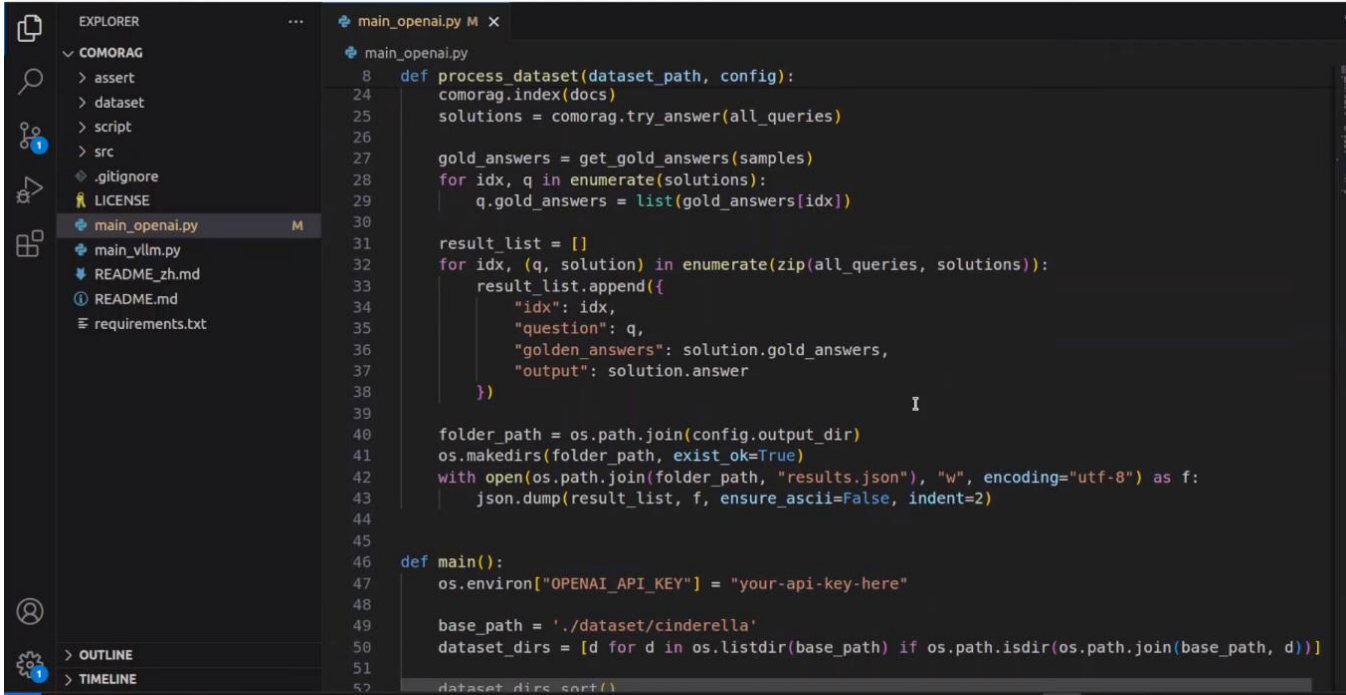
```
Ubuntu@0360-kci2-ty6k-prxmx100037: ~/ComoRAG
Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#   $ conda activate ai
#
# To deactivate an active environment, use
#
#   $ conda deactivate

(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~$ git clone https://github.com/EternityJune25/ComoRAG.git && cd ComoRAG
Cloning into 'ComoRAG'...
remote: Enumerating objects: 97, done.
remote: Counting objects: 100% (97/97), done.
remote: Compressing objects: 100% (74/74), done.
remote: Total 97 (delta 20), reused 91 (delta 15), pack-reused 0 (from 0)
Receiving objects: 100% (97/97), 1.24 MiB | 7.45 MiB/s, done.
Resolving deltas: 100% (20/20), done.
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/ComoRAG$ pip install -r requirements.txt
Collecting accelerate==1.6.0 (from -r requirements.txt (line 1))
  Downloading accelerate-1.6.0-py3-none-any.whl.metadata (19 kB)
Collecting aiohappyeyeballs==2.6.1 (from -r requirements.txt (line 2))
  Using cached aiohappyeyeballs-2.6.1-py3-none-any.whl.metadata (5.9 kB)
```

```
ece-0.2.0 sentry-sdk-2.27.0 setproctitle-1.3.5 shellingham-1.5.4 six-1.17.0 smart-open-7.1.0 smmap-5.0.2 starlette-0.46.2 tenacity-8.2.3 texttable-1.7.0 threadpoolctl-3.6.0 tiktoken-0.9.0 tokenizers-0.21.1 torch-2.6.0 torchaudio-2.6.0 torchvision-0.21.0 transformers-4.51.3 triton-3.2.0 typer-0.15.3 typing_extensions-4.13.2 umap-learn-0.5.5 urllib3-2.5.0 uvicorn-0.34.2 uvloop-0.21.0 virtualenv-20.30.0 vllm-0.8.5.post1 vpd-1.0.0 wandb-0.19.10 watchfiles-1.0.5 websockets-15.0.1 wrapt-1.17.2 xformers-0.0.29.post2 xgrammar-0.1.18 xxhash-3.5.0 yarl-1.20.0 zip
(ai) Ubuntu@0360-kci2-ty6k-prxmx100037:~/ComoRAG$ export OPENAI_API_KEY=
```

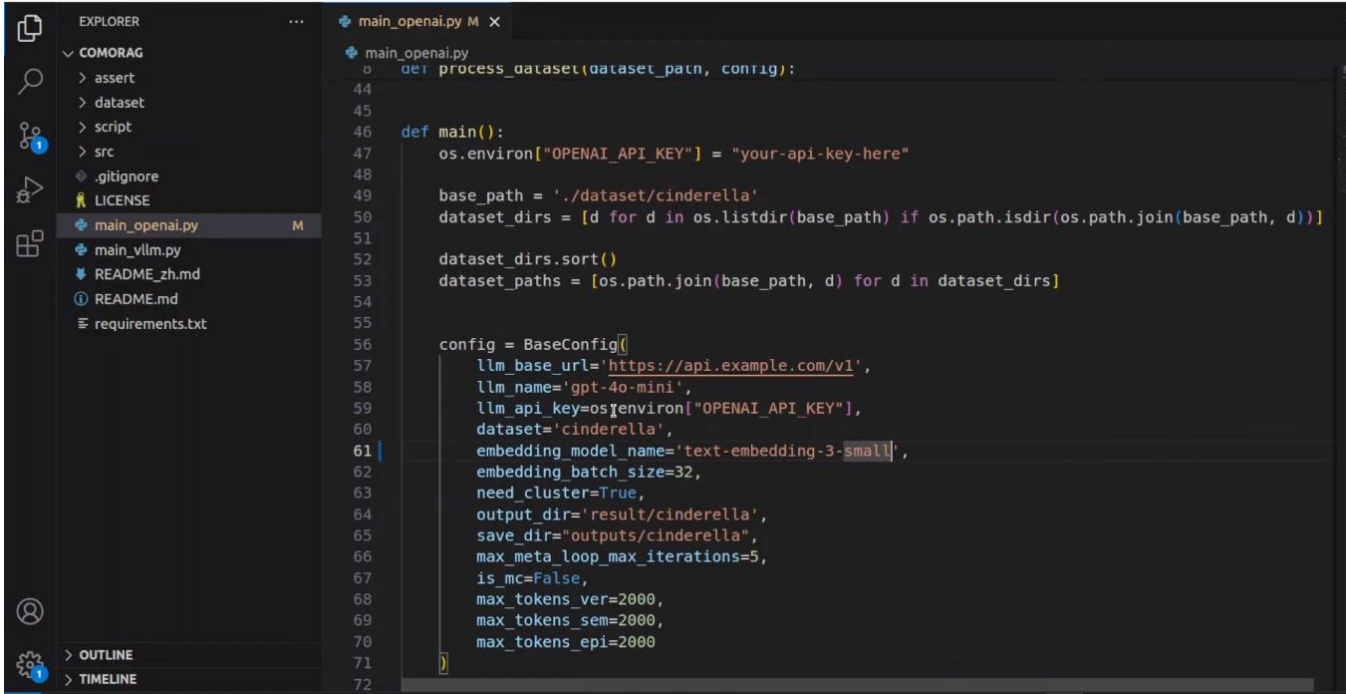
```
FILE  NAME  DESCRIPTION  VIEW  NEW  FAVORITES  HELP
main_openai.py M X
main_openai.py
1  import os
2  import json
3  from src.comorag.ComoRAG import ComoRAG
4  from src.comorag.utils.config_utils import BaseConfig
5  from src.comorag.utils.misc_utils import get_gold_answers
6
7
8  def process_dataset(dataset_path, config):
9      dataset_name = os.path.basename(dataset_path)
10     corpus_path = os.path.join(dataset_path, "corpus.jsonl")
11     qas_path = os.path.join(dataset_path, "qas.jsonl")
12
13     with open(corpus_path, 'r', encoding='utf-8') as f:
14         corpus = [json.loads(line) for line in f if line.strip()]
15     docs = [doc['contents'] for doc in corpus]
16
17     with open(qas_path, 'r', encoding='utf-8') as f:
18         samples = [json.loads(line) for line in f if line.strip()]
19
20     all_queries = [s['question'] for s in samples]
21     config.corpus_len=len(corpus),
22
23     comorag = ComoRAG(global_config=config)
24     comorag.index(docs)
25     solutions = comorag.try_answer(all_queries)
26
27     gold_answers = get_gold_answers(samples)
28     for idx, q in enumerate(solutions):
29         q.gold_answers = list(gold_answers[idx])
30
```



```

8  def process_dataset(dataset_path, config):
24     comorag.index(docs)
25     solutions = comorag.try_answer(all_queries)
26
27     gold_answers = get_gold_answers(samples)
28     for idx, q in enumerate(solutions):
29         q.gold_answers = list(gold_answers[idx])
30
31     result_list = []
32     for idx, (q, solution) in enumerate(zip(all_queries, solutions)):
33         result_list.append({
34             "idx": idx,
35             "question": q,
36             "golden_answers": solution.gold_answers,
37             "output": solution.answer
38         })
39
40     folder_path = os.path.join(config.output_dir)
41     os.makedirs(folder_path, exist_ok=True)
42     with open(os.path.join(folder_path, "results.json"), "w", encoding="utf-8") as f:
43         json.dump(result_list, f, ensure_ascii=False, indent=2)
44
45
46 def main():
47     os.environ["OPENAI_API_KEY"] = "your-api-key-here"
48
49     base_path = './dataset/cinderella'
50     dataset_dirs = [d for d in os.listdir(base_path) if os.path.isdir(os.path.join(base_path, d))]
51
52     dataset_dirs.sort()

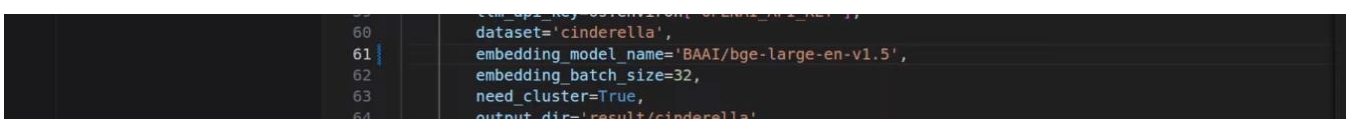
```



```

44     def process_dataset(dataset_path, config):
45
46     def main():
47         os.environ["OPENAI_API_KEY"] = "your-api-key-here"
48
49         base_path = './dataset/cinderella'
50         dataset_dirs = [d for d in os.listdir(base_path) if os.path.isdir(os.path.join(base_path, d))]
51
52         dataset_dirs.sort()
53         dataset_paths = [os.path.join(base_path, d) for d in dataset_dirs]
54
55
56         config = BaseConfig({
57             llm_base_url='https://api.example.com/v1',
58             llm_name='gpt-4o-mini',
59             llm_api_key=os.getenv("OPENAI_API_KEY"),
60             dataset='cinderella',
61             embedding_model_name='text-embedding-3-small',
62             embedding_batch_size=32,
63             need_cluster=True,
64             output_dir='result/cinderella',
65             save_dir="outputs/cinderella",
66             max_meta_loop_max_iterations=5,
67             is_mc=False,
68             max_tokens_ver=2000,
69             max_tokens_sem=2000,
70             max_tokens_epi=2000
71         })
72

```



```

60     llm_api_key=os.getenv("OPENAI_API_KEY"),
61     dataset='cinderella',
62     embedding_model_name='BAAI/bge-large-en-v1.5',
63     embedding_batch_size=32,
64     need_cluster=True,
65     output_dir='result/cinderella'

```

```
File Edit Selection View Go Run Terminal Help

EXPLORER
COMORAG
  > assert
  > dataset
  > script
  > src/comorag
    > embedding_model
    > information_extraction
    > llm
    > prompts
    > utils
  > _init_.py
  > ComoRAG.py
  > embedding_store.py
  > rerank.py
  > .gitignore
  > LICENSE
  > main_openai.py
  > main_vllm.py
  > README_zh.md
  > README.md
  > requirements.txt

main_openai.py M
ComoRAG.py X

src > comorag > ComoRAG.py
1 from gc import collect
2 import json
3 from multiprocessing import pool
4 import os
5 import logging
6 from dataclasses import dataclass, field, asdict
7 from datetime import datetime
8 from typing import Union, Optional, List, Set, Dict, Any, Tuple, Literal
9 from anyio import Semaphore
10 from click import prompt
11 from cv2 import log
12 import numpy as np
13 from collections import defaultdict
14 from tqdm import tqdm
15 import igraph as ig
16 import re
17 import concurrent.futures
18 from transformers import AutoTokenizer
19 from concurrent.futures import ThreadPoolExecutor, as_completed
20
21 from .utils.embed_utils import get_similar_summaries
22 from .utils import agents
23 from src.comorag.utils.timeline_utils import TimelineSummarizer
24 from src.comorag.utils.summarization_utils import GPT4SummarizationModel
25
26 from .llm import get_llm_class, BaseLLM
27 from .embedding_model import get_embedding_model_class, BaseEmbeddingModel
28 from .embedding_store import EmbeddingStore
29 from .information_extraction import OpenIE
30 from .information_extraction.openai_vllm_offline import VLLMOfflineOpenIE
```

```
EXPLORER
COMORAG
  > assert
  > dataset/cinderella/cinderella_1
    > cinderella.txt
    > corpus.jsonl
    > qas.jsonl
  > script
  > src/comorag
    > .gitignore
    > LICENSE
    > main_openai.py
    > main_vllm.py
    > README_zh.md
    > README.md
    > requirements.txt

main_openai.py M
cinderella.txt X

dataset > cinderella > cinderella_1 > cinderella.txt
1 The wife of a rich man fell sick, and as she felt that her end
2 was drawing near, she called her only daughter to her bedside and
3 said, dear child, be good and pious, and then the
4 good God will always protect you, and I will look down on you
5 from heaven and be near you. Thereupon she closed her eyes and
6 departed. Every day the maiden went out to her mother's grave,
7 and wept, and she remained pious and good. When winter came
8 the snow spread a white sheet over the grave, and by the time the
9 spring sun had drawn it off again, the man had taken another wife.
10 The woman had brought with her into the house two daughters,
11 who were beautiful and fair of face, but vile and black of heart.
12 Now began a bad time for the poor step-child. Is the stupid goose
13 to sit in the parlor with us, they said. He who wants to eat bread
14 must earn it. Out with the kitchen-wench. They took her pretty
15 clothes away from her, put an old grey bedgown on her, and gave
16 her wooden shoes. Just look at the proud princess, how decked
17 out she is, they cried, and laughed, and led her into the kitchen.
18 There she had to do hard work from morning till night, get up
19 before daybreak, carry water, light fires, cook and wash. Besides
20 this, the sisters did her every imaginable injury - they mocked her
21 and emptied her peas and lentils into the ashes, so that she was
22 forced to sit and pick them out again. In the evening when she had
23 worked till she was weary she had no bed to go to, but had to sleep
24 by the hearth in the cinders. And as on that account she always
```

```
EXPLORER
COMORAG
  > assert
  > dataset/cinderella/cinderella_1
    > cinderella.txt
    > corpus.jsonl
    > qas.jsonl
  > script
  > src/comorag
    > .gitignore
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    > main_openai.py
    > main_vllm.py
    > README_zh.md
    > README.md
    > requirements.txt

main_openai.py M
cinderella.txt X

dataset > cinderella > cinderella_1 > cinderella.txt
58 two hours, you shall go with us. The maiden went through the
59 back-door into the garden, and called, you tame pigeons, you
60 turtle-doves, and all you birds beneath the sky, come and help me
61 to pick
62 the good into the pot,
63 the bad into the crop.
64 Then two white pigeons came in by the kitchen window, and
65 afterwards the turtle-doves, and at last all the birds beneath the
66 sky, came whirring and crowding in, and alighted amongst the ashes.
67 And the pigeons nodded with their heads and began pick, pick,
68 pick, pick, and the rest began also pick, pick, pick, pick, and
69 gathered all the good grains into the dish. Hardly had one hour
70 passed before they had finished, and all flew out again. Then the
71 girl took the dish to her step-mother, and was glad, and believed
72 that now she would be allowed to go with them to the festival.
73 But the step-mother said, no, cinderella, you have no clothes and
74 you can not dance. You would only be laughed at. And as
75 cinderella wept at this, the step-mother said, if you can pick two
76 dishes of lentils out of the ashes for me in one hour, you shall go
77 with us. And she thought to herself, that she most certainly
78 cannot do again. When the step-mother had emptied the two
79 dishes of lentils amongst the ashes, the maiden went through the
80 back-door into the garden and cried, you tame pigeons, you
81 turtle-doves, and all you birds beneath the sky, come and help me
82 to pick
83 the good into the pot,
84 the bad into the crop.
85 Then two white pigeons came in by the kitchen-window, and
86 afterwards the turtle-doves, and at length all the birds beneath the
87 sky, came whirring and crowding in, and alighted amongst the
```


Home / ComoRAG / outputs / cinderella / cinderella_1 / gpt-4o-mini_BAAI_bge-m3			
	Name	Size	Modified
Recent	chunk_embeddings	1 item	00:41 ☆
Starred	entity_embeddings	0 items	00:41 ☆
Home	fact_embeddings	0 items	00:41 ☆
Desktop	summary_embeddings	1 item	00:41 ☆
Documents	timeline_embeddings	4 items	00:41 ☆
Downloads	final_summary.txt	369 bytes	00:45 ☆
Music			
Pictures			
Videos			
Trash			

```

1 What did Cinderella's mother tell her before she died?
2 She told Cinderella to be good and pious, and that God would protect her.
3
4 How did the prince find Cinderella?
5 By using the golden slipper—the shoe fit only Cinderella.
6
7 What happens when the snow covers Cinderella's mother's grave?
8 Cinderella's father marries again and the stepmother comes into the house.

```

```

File Edit Selection View Go Run Terminal Help
EXPLORER
COMORAG
  > assert
  > dataset/cinderella/cinderella_1
    cinderella.txt
    corpus.jsonl
    qas.jsonl
  > outputs
  > script
  > src
    .gitignore
    cuda-keyring_1.1-1_all.deb U
    cudnn-local-repo-ubuntu22... U
    LICENSE
    main_openai.py M
    main_vllm.py
    README_zh.md
    README.md
    requirements.txt
  > OUTLINE
  > TIMELINE

dataset > cinderella > cinderella_1 > cinderella.txt
42 played, and a little white bird always came on the tree, and at
43 cinderella expressed a wish, the bird threw down to her what she
44 had wished for.
45 It happened, however, that the king gave orders for a festival
46 which was to last three days, and to which all the beautiful young
47 girls in the country were invited, in order that his son might choose
48 himself a bride. When the two step-sisters heard that they too were
49 to appear among the number, they were delighted, called cinderella
50 and said, comb our hair for us, brush our shoes and fasten our
51 buckles, for we are going to the wedding at the king's palace.
52 Cinderella obeyed, but wept, because she too would have liked to
53 go with them to the dance, and begged her step-mother to allow
54 her to do so. You go, cinderella, said she, covered in dust and
55 dirt as you are, and would go to the festival. You have no clothes
56 and shoes, and yet would dance. As, however, cinderella went on
57 asking, the step-mother said at last, I have emptied a dish of
58 lentils into the ashes for you, if you have picked them out again in
59 two hours, you shall go with us. The maiden went through the
60 back-door into the garden, and called, you tame pigeons, you
61 turtle-doves, and all you birds beneath the sky, come and help me
62 to pick
63 the good into the pot,
64 the bad into the crop.
65 Then two white pigeons came in by the kitchen window, and
66 afterwards the turtle-doves, and at last all the birds beneath the
67 sky, came whirring and crowding in, and alighted amongst the ashes.
68 And the pigeons nodded with their heads and began pick, pick,
69 pick, pick, and the rest began also pick, pick, pick, pick, and
70 gathered all the good grains into the dish. Hardly had one hour
    passed before they had finished, and all flew out again. Then the

```

```

File Edit Selection View Go Run Terminal Help
EXPLORER
COMORAG
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    cinderella.txt
    corpus.jsonl
    qas.jsonl
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    cudnn-local-repo-ubuntu22... U
    LICENSE
    main_openai.py M
    main_vllm.py
    README_zh.md
    README.md
    requirements.txt

dataset > cinderella > cinderella_1 > qas.jsonl
1 {"id": "1", "question": "What did Cinderella's mother tell her before she died?", "golden_answers":
2 {"id": "2", "question": "How did the prince find Cinderella?", "golden_answers": ["Golden slipper",
3 {"id": "3", "question": "What happens when the snow covers Cinderella's mother's grave?", "golden a
4
5

```